ABRAR HOSSAIN

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Education

The University of Toledo — Toledo, Ohio
PhD, Computer Science, GPA 3.87/4.00

The University of Toledo — Toledo, Ohio
Master's, Computer Science, GPA 3.90/4.00 — August 2023 – August 2025

Chittagong University of Engineering and Technology — Chittagong, Bangladesh
Bachelors, Electrical Engineering, GPA 3.23/4.00 — March 2015 – September 2019

Experience

Graduate Research Assistant — August 2023 – Present
The University of Toledo — Toledo, OH

- Developed TARDIS, a power-aware HPC scheduler using GNNs, cost reduction: 18% (temporal), 10-20% (spatial)
- Developed LASP, a MAB-based HPC tuner on edge devices, achieving 2.5% average performance gain over default
- Developed LoopRL, a reinforcement-learning MLIR loop transformation framework achieving up to 32.1× speedup

Intern — June 2025 – August 2025
National Center for Supercomputing Applications — Urbana, IL

- Built parser for eBPF maps for 80+ VLANs and ports with configurable polling intervals via REST API
- Integrated with InfluxDB for efficient storage and Grafana for real-time visualization
- Added configurable logging, reducing manual reconfiguration time by 40%

Visitor — August 2024 – December 2024
NSF National Center for Atmospheric Research — Remote

- Set up CouchDB, Chords, and Streampipes on ACCESS Jetstreams for community weather data storage
- Built data orchestrator for efficient data routing, achieving 39% transmission efficiency gain
- Reduced deployment costs by 22% for communities implementing the project

Intern — May 2024 – August 2024
NSF National Center for Atmospheric Research — Boulder, CO

- Designed private LoRa network for 6+ data types with Raspberry Pi gateways and central server.
- Improved wind forecasting with edge-ML, achieving 26% accuracy gain on Raspberry Pi
- Image analysis with TensorFlow, 23% accuracy gain, 3x faster training, and 93% precision on 10,000+ images

Publications

[Paper] • **(PASC 2026)** • **Abrar Hossain**, Kishwar Ahmed **XOR Bidding and Knapsack Formulations for HPC Network Resource Allocation**

[Paper] • **(JSSPP 2025)** • **Abrar Hossain**, Abubeker Abdurahman, Mohammad Atiqul Islam, Kishwar Ahmed. **Power-Aware Scheduling for Multi-Center HPC Electricity Cost Optimization**

[Paper] • **(HiPC 2024)** • **Abrar Hossain**, Abdel-Hameed Badawy, Mohammad Atiqul Islam, Tapasya Patki, Kishwar Ahmed. **HPC Application Parameter Autotuning on Edge Devices: A Bandit Learning Approach**

[Paper] • **(EuroSys 2026)** • Md. Azahar Alam, **Abrar Hossain**, Kishwar Ahmed. **Dynamic Instrumentation for Dataflow Accelerators**

[Paper] • **(WSC 2024)** • Abubeker Abdurahman, **Abrar Hossain**, Kevin A Brown, Kazutomo Yoshii, Kishwar Ahmed. **Scalable HPC Job Scheduling and Resource Management in SST**

[Paper] • **(IoTBDS 2025)** • Tasnimul Hasan, **Abrar Hossain**, Mufakir Qamar Ansari, Talha Hussain Syed **Enhanced Intrusion Detection in IIoT Networks: A Lightweight Approach with Autoencoder-Based Feature Learning**

[Paper (Under Review)] • **(CCGrid 2027)** • **Abrar Hossain**, Kishwar Ahmed. **Structured Reinforcement Learning for Loop Transformation in MLIR**

[Paper (Under Review)] • **(SIGMETRICS 2027)** • **Abrar Hossain**, Kishwar Ahmed. **Universal Acquisition for Risk Certification with Informative Completion Times**

[Paper (Under Review)] • **(PPoPP 2027)** • Faeghe Farrokhpey, **Abrar Hossain**, Kishwar Ahmed. **CAWS: Congestion-Aware Warp Scheduling for GPU Memory-NoC Bottlenecks**

[Journal Paper (Under Review)] • **(ACM TOPC)** • **Abrar Hossain**, Xingfu Wu, Jason Liu, Tapasya Patki, Kishwar Ahmed. **Autotuning Across the High-Performance Computing Stack: Hardware, Software, and Application Optimization Techniques**

[Workshop Paper] • **(EduHiPC 2024)** • Abubeker Abdurahman, Arihant Singh, **Abrar Hossain**, Kishwar Ahmed. **A Hands-On Approach To Teaching Parallel and Heterogeneous Computing**

[Book Chapter] • **(TPCPD 2025)** • Abubeker Abdurahman, **Abrar Hossain**, Kishwar Ahmed. **An Interactive Learning Module for Introducing Parallel Computing**

[Poster] • **(SC 2025)** • **Abrar Hossain**, Kishwar Ahmed. **CROSS-HPC System Bayesian Optimization with Adaptive Transfer**

[Poster] • **(eScience 2025)** • **Abrar Hossain**, Kishwar Ahmed. **Bandwidth Allocation for Heterogeneous HPC Data Ingestion using Dynamic Auctions**

[Poster] • **(AGU 2024)** • **Abrar Hossain**, Keith Maull, Agbeli Ameko. **Environmental Data Sensing and Monitoring System Using Community-based Private LoRa Network**

[Poster] • **(SC 2023)** • **Abrar Hossain**, Kishwar Ahmed. **Automating HPC Model Selection on Edge Devices**

Awards & Honors

SGX3 Rising Star of the Year Award (2025)

University of Toledo Graduate Research Assistant Recognition Award (2025)

Midwest RCD Symposium Scholarship (2025)

CRA-WP Grad Cohort for IDEALS Scholarship (2024)

Friends Education Fund Scholarship (2024)

Bangladesh Sweden Trust Fund Scholarship (2024)

Research & Travel Grants

NSF Travel Grants: eScience 2025, Cluster 2025

Workshop Travel Grant: BRICCS Pathway'26, RMACC HPC Sys Admin'26, CI4FAIR'25, Midwest RCD('25,'26). RCD Nexus'26

SIParCS Student Travel Grant (2024, 2025)

USRSE25 Building Engagement (BE) Travel Grant (2025)

AGU Student Travel Grant (2024)

IEEE HiPC TCPP Travel Grant (2024)

SIGHPC PASC Travel Grant (2026)

Professional and Volunteer Activities

Professional Engagements:

- Student Champion for *NSF ACCESS*
- Reviewer for *Future Generation Computer Systems*
- Reviewer for *HPEC'25 – Extended Abstracts*
- PC Member for *PEARC'26 – Tutorials and Workshops*
- PC Member for *PEARC'25 – Short Papers, Tutorials and Workshops*
- PC Member for *PEARC'25 – Posters and Visualization*

Conference Volunteer Roles:

- Student Volunteer at *PASC'26 – Platform for Advanced Scientific Computing*
- Student Volunteer at *PEARC'25 – Practice and Experience in Advanced Research Computing*
- Student Volunteer at *HiPC'24 – International Conference on High Performance Computing, Data, and Analytics*
- Student Volunteer at *AGU'24 – American Geophysical Union Fall Meeting*

Technical Skills

Programming Languages: Python, C, C++, R
Operating Systems: Linux (Ubuntu, CentOS, Debian), Windows Server (Active Directory, Group Policy)
Networking: TCP/IP, DNS, DHCP, Firewalls (iptables, ufw), VPN (WireGuard, OpenVPN)
Automation & Scripting: Bash, Python, PowerShell, Ansible, Terraform
Cloud Platforms: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure
Virtualization & Containers: VMware, KVM, Docker, Kubernetes, Proxmox
Monitoring & Logging: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Nagios
Backup & Disaster Recovery: Rsync, Bacula, Veeam, RAID
Version Control & CI/CD: Git, GitHub, GitLab, Jenkins, ArgoCD